

Multiple Choice

1. **Answer:** A

Options: A) $1.256 \times 10^{-46} \text{kgm}^2$ and $9.100 \times 10^{-2} \text{nm}$; B) $1.336 \times 10^{-47} \text{kgm}^2$ and $8.250 \times 10^{-2} \text{nm}$; C) $1.336 \times 10^{-46} \text{kgm}^2$ and $9.196 \times 10^{-1} \text{nm}$; D) $1.410 \times 10^{-47} \text{kgm}^2$ and $9.250 \times 10^{-2} \text{nm}$; E) $1.256 \times 10^{-47} \text{kgm}^2$ and $8.196 \times 10^{-2} \text{nm}$

Note: Correct option : A - $1.256 \times 10^{-46} \text{kgm}^2$ and $9.100 \times 10^{-2} \text{nm}$

1. **Answer:** E

Options: A) $4 \times 10^{-15} \text{J}$; B) $1.2 \times 10^{-15} \text{J}$; C) $5 \times 10^{-15} \text{J}$; D) $3.5 \times 10^{-15} \text{J}$; E) 210^{-15}J

Note: Correct option: E - $2 \times 10^{-15} \text{J}$

2. **Answer:** A

Options: A) the protons, electrons and neutrons; B) the electrons and protons; C) the nucleus; D) the protons; E) the neutrons

Note: Correct option: A - the protons, electrons and neutrons

3. **Answer:** A

Options: A) -2; B) 1; C) -0.5; D) 0.5; E) 2

Note: Correct option: A - -2

4. **Answer:** A

Options: A) 0.11 J/g ; B) 0.20 J/g ; C) 0.13 J/g ; D) 0.18 J/g ; E) 0.30 J/g *Note:* Correct option: A - 0.11 J/g

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '7'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'NH₂'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '28 electrons, 27 protons, 24 neutrons'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'SnF₄, PH₃, Si₂O₅'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Both 35 Cl and 37 Cl have electric quadrupole moments.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 72.42% and 27.58%. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key